

Mini Project 1:

“Student Productivity Tracker” Objective: Combine loops, arrays, and file handling to build a console-based app that logs daily study hours and generates a weekly progress report. Trainer Coverage: Guide students to design menu-driven interface. Discuss file I/O for persistence.

Tasks

1. Create a console-based app called Music Listening Logger that allows users to enter the number of minutes they listened to music each day for a week and stores this data in an array.

```
1 #include<stdio.h>
2 int main(){
3     int music[7];
4     printf("Music listening logger");
5     for(int i=0;i<7;i++){
6         printf("Enter music listening minutes for day %d:",i+1);
7         scanf("%d",&music[i]);
8     }
9     printf("Weekly music listening");
10    for(int i=0;i<7;i++){
11        printf("\nDay %d:%d minutes\n",i+1,music[i]);
12    }
13    return 0;
14 }
```

Output:

```
Music listening logger
Enter music listening minutes for day 1:2
Enter music listening minutes for day 2:3
Enter music listening minutes for day 3:1
Enter music listening minutes for day 4:3
Enter music listening minutes for day 5:4
Enter music listening minutes for day 6:2
Enter music listening minutes for day 7:5
Weekly music listening
Day 1:2 minutes
Day 2:3 minutes
Day 3:1 minutes
Day 4:3 minutes
Day 5:4 minutes
Day 6:2 minutes
Day 7:5 minutes
-----
Process exited after 6.722 seconds with return value 0
Press any key to continue . . . |
```

2. Add a menu-driven interface to your Music Listening Logger so users can choose to log new listening minutes, view their weekly summary, or exit the app.

Hint: Use a loop to repeatedly show the menu until the user selects exit.

```
1  #include <stdio.h>
2  int main(){
3      int choice;
4      int music[7] = {0};
5      do{
6          printf("MUSIC LISTENING LOGGER\n");
7          printf("1. Log New Listening Minutes\n");
8          printf("2. View Weekly Summary\n");
9          printf("3. Exit\n");
10
11         printf("\nEnter your choice: ");
12         scanf("%d", &choice);
13
14         if (choice == 1){
15             printf("\n----- ENTER MUSIC MINUTES ----- \n");
16             for (int i = 0; i < 7; i++){
17                 printf("Day %d: ", i + 1);
18                 scanf("%d", &music[i]);
19             }
20             printf("\nMusic listening minutes saved successfully!\n");
21         }
22
23         else if (choice == 2){
24             printf("\n----- WEEKLY MUSIC SUMMARY ----- \n");
25             for (int i = 0; i < 7; i++){
26                 printf("Day %d : %d minutes\n",
27                     i + 1, music[i]);
28             }
29         }
30
31         else if (choice == 3){
32             printf("\nThank you for using Music Listening Logger!\n");
33         }
34
35         else{
36             printf("\nInvalid choice! Please try again.\n");
37         }
38     }
39     while (choice != 3);
40     return 0;
41 }
```

Output:

```
MUSIC LISTENING LOGGER
1. Log New Listening Minutes
2. View Weekly Summary
3. Exit

Enter your choice: 1

----- ENTER MUSIC MINUTES -----
Day 1: 3
Day 2: 5
Day 3: 3
Day 4: 5
Day 5: 6
Day 6: 2
Day 7: 8

Music listening minutes saved successfully!
MUSIC LISTENING LOGGER
1. Log New Listening Minutes
2. View Weekly Summary
3. Exit

Enter your choice: 2

----- WEEKLY MUSIC SUMMARY -----
Day 1 : 3 minutes
Day 2 : 5 minutes
Day 3 : 3 minutes
Day 4 : 5 minutes
Day 5 : 6 minutes
Day 6 : 2 minutes
Day 7 : 8 minutes
MUSIC LISTENING LOGGER
1. Log New Listening Minutes
2. View Weekly Summary
3. Exit

Enter your choice: 3

Thank you for using Music Listening Logger!

-----
Process exited after 21.25 seconds with return value 0
Press any key to continue . . . |
```

3. Implement file handling in your Music Listening Logger so that when the user logs their daily listening minutes, the data is saved to a file named music_log.txt for persistence.

```
1  #include <stdio.h>
2  int main(){
3      FILE *fp;
4      int minutes[7];
5      printf("MUSIC LISTENING LOGGER\n\n");
6
7      for(int i = 0; i < 7; i++){
8          printf("Enter listening minutes for Day %d: ", i + 1);
9          scanf("%d", &minutes[i]);
10     }
11
12     fp = fopen("music_log.txt", "w");
13
14     if(fp == NULL){
15         printf("\nFile could not be created.\n");
16         return 1;
17     }
18
19     for(i = 0; i < 7; i++){
20         fprintf(fp, "Day %d : %d minutes\n",
21             i + 1, minutes[i]);
22     }
23
24     fclose(fp);
25     printf("\nData saved successfully to music_log.txt\n");
26     return 0;
27 }
```

Output:

```
D:\06Aug_SE\C practice\Mini: x + v
MUSIC LISTENING LOGGER

Enter listening minutes for Day 1: 4
Enter listening minutes for Day 2: 3
Enter listening minutes for Day 3: 2
Enter listening minutes for Day 4: 5
Enter listening minutes for Day 5: 6
Enter listening minutes for Day 6: 8
Enter listening minutes for Day 7: 7

Data saved successfully to music_log.txt

-----
Process exited after 9.789 seconds with return value 0
Press any key to continue . . . |
```

4. Add a feature to your app that reads the saved music_log.txt file and generates a weekly report showing the total, average, and highest listening minutes in the week.

```
1  #include <stdio.h>
2  int main(){
3      FILE *fp;
4      int minutes;
5      int total = 0;
6      int highest = 0;
7      float average;
8      fp = fopen("music_log.txt", "r");
9
10     if (fp == NULL){
11         printf("File not found!\n");
12         return 1;
13     }
14     printf("WEEKLY MUSIC REPORT\n\n");
15     for (i = 1; i <= 7; i++){
16         fscanf(fp, "Day %d : %d minutes", &minutes);
17         printf("Day %d : %d minutes\n", i, minutes);
18         total = total + minutes;
19
20         if (minutes > highest){
21             highest = minutes;
22         }
23     }
24     fclose(fp);
25     average = total / 7.0;
26
27     printf("Total    : %d minutes\n", total);
28     printf("Average : %.2f minutes\n", average);
29     printf("Highest : %d minutes\n", highest);
30
31     return 0;
32 }
```

Output:

```
D:\06Aug_SE\C practice\Mini4 × + v
WEEKLY MUSIC REPORT
Day 1 : 4 minutes
Day 2 : 4 minutes
Day 3 : 4 minutes
Day 4 : 4 minutes
Day 5 : 4 minutes
Day 6 : 4 minutes
Day 7 : 4 minutes
Total : 28 minutes
Average : 4.00 minutes
Highest : 4 minutes

-----
Process exited after 0.09787 seconds with return value 0
Press any key to continue . . . |
```

5. Refactor your Music Listening Logger to allow users to reset their weekly data by clearing the array and deleting the contents of music_log.txt.

Constraint: Make sure the reset option is available in the menu and asks for confirmation before deleting data.

```

1  #include <stdio.h>
2  int main(){
3      int music[7] = {0};
4      int choice;
5      int confirm;
6      do{
7          printf("MUSIC LISTENING LOGGER\n");
8          printf("1. Log New Listening Minutes\n");
9          printf("2. View Weekly Summary\n");
10         printf("3. Reset Weekly Data\n");
11         printf("4. Exit\n");
12
13         printf("\nEnter your choice: ");
14         scanf("%d", &choice);
15
16         if (choice == 1){
17             printf("\n----- ENTER MUSIC MINUTES ----- \n");
18             for (int i = 0; i < 7; i++){
19                 printf("Enter listening minutes for Day %d: ", i + 1);
20                 scanf("%d", &music[i]);
21             }
22
23             FILE *fp;
24             fp = fopen("music_log.txt", "w");
25
26             if (fp == NULL){
27                 printf("File could not be opened!\n");
28             }
29             else{
30                 for (i = 0; i < 7; i++){
31                     fprintf(fp, "Day %d : %d minutes\n",
32                             i + 1, music[i]);
33                 }
34                 fclose(fp);
35                 printf("\nData saved successfully!\n");
36             }
37         }
38
39         else if (choice == 2){
40             FILE *fp;
41             int minutes;
42             int total = 0;
43             int highest = 0;
44             float average;
45
46             fp = fopen("music_log.txt", "r");
47
48             if (fp == NULL){
49                 printf("\nNo music data found!\n");
50             }
51             else{
52                 printf("\n----- WEEKLY MUSIC SUMMARY ----- \n");
53
54                 for (i = 1; i <= 7; i++){

```

```

54 for (i = 1; i <= 7; i++){
55     fscanf(fp, "Day %d : %d minutes",
56             &minutes);
57
58     printf("Day %d : %d minutes\n",
59            i, minutes);
60
61     total = total + minutes;
62     if (minutes > highest){
63         highest = minutes;
64     }
65 }
66
67 fclose(fp);
68 average = total / 7.0;
69
70 printf("\nTotal    : %d minutes\n", total);
71 printf("Average : %.2f minutes\n", average);
72 printf("Highest  : %d minutes\n", highest);
73 }
74 }
75
76 /* OPTION 3: Reset data */
77 else if (choice == 3)
78 {
79     printf("\nWARNING: This will delete all weekly data.\n");
80     printf("Are you sure you want to reset? (1 = Yes, 0 = No): ");
81     scanf("%d", &confirm);
82
83     if (confirm == 1)
84     {
85         /* Clear the array */
86         for (i = 0; i < 7; i++)
87         {
88             music[i] = 0;
89         }
90
91         /* Clear the file */
92         FILE *fp;
93         fp = fopen("music_log.txt", "w");
94
95         if (fp != NULL)
96         {
97             fclose(fp);
98
99             printf("\nWeekly data has been reset successfully!\n");
100         }
101         else
102         {
103             printf("\nFile could not be cleared!\n");
104         }
105     }
106 }

```

```

105         else
106         {
107             printf("\nReset cancelled. Your data is safe.\n");
108         }
109     }
110
111     /* OPTION 4: Exit */
112     else if (choice == 4)
113     {
114         printf("\nThank you for using Music Listening Logger!\n");
115     }
116
117     else
118     {
119         printf("\nInvalid choice! Please try again.\n");
120     }
121
122     } while (choice != 4);
123
124     return 0;
125 }

```

Output:

```

MUSIC LISTENING LOGGER
1. Log New Listening Minutes
2. View Weekly Summary
3. Reset Weekly Data
4. Exit

Enter your choice: 1

----- ENTER MUSIC MINUTES -----
Enter listening minutes for Day 1: 3
Enter listening minutes for Day 2: 6
Enter listening minutes for Day 3: 4
Enter listening minutes for Day 4: 5
Enter listening minutes for Day 5: 2
Enter listening minutes for Day 6: 1
Enter listening minutes for Day 7: 7

Data saved successfully!
MUSIC LISTENING LOGGER
1. Log New Listening Minutes
2. View Weekly Summary
3. Reset Weekly Data
4. Exit

Enter your choice: 2

----- WEEKLY MUSIC SUMMARY -----
Day 1 : 3 minutes
Day 2 : 3 minutes
Day 3 : 3 minutes
Day 4 : 3 minutes
Day 5 : 3 minutes
Day 6 : 3 minutes
Day 7 : 3 minutes

Total : 21 minutes
Average : 3.00 minutes
Highest : 3 minutes
MUSIC LISTENING LOGGER
1. Log New Listening Minutes
2. View Weekly Summary
3. Reset Weekly Data
4. Exit

```